



MODULE NAME:	MODULE CODE:
PROGRAMMING 1B	PROG6112

ASSESSMENT TYPE: PRACTICAL ASSIGNMENT 1 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 15 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Your assignment must be submitted through Turnitin.***
3. ***Save a copy of your assignment before submitting it.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***All work must be adequately and correctly referenced.***
6. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Assignment Instructions

1. Each question indicates a sample result. This should be used as a guide to the expected format and/or result, providing you with a clear understanding of what is expected. However, in some cases, the sample result may only include a single record of the expected results.
2. For mark allocation, it is crucial that you refer to the Marking Rubrics at the end of each question. These rubrics provide a clear breakdown of how your work will be assessed.
3. *Do not submit AI-generated code as your own work. Submitting AI code without your analysis, synthesis, and critical thinking may be considered plagiarism or academic misconduct.*
4. *If you used AI tools in any part of your process (e.g., planning, proofreading, coding), it's advisable to disclose their use briefly in a footnote or comments section (e.g., "ChatGPT was consulted for initial topic brainstorming only").*

Section A**(Marks: 65)****Question 1****(Marks: 40)**

A new TV streaming company has entered the market and requires a Java application to keep track of the latest TV series details on offer to customers. The streaming company has recently opened a branch in the city where you reside. It has hired the software development house you work for to design a Java application to manage their streaming options.

Your line manager has requested you to develop the TV series management application with the following requirements:

1.1. When the application starts, it must display the following menu structure:

Sample Menu Screen Shot

```
LATEST SERIES - 2025
*****
Enter (1) to launch menu or any other key to exit
```

```
Please select one of the following menu items:
(1) Capture a new series.
(2) Search for a series.
(3) Update series age restriction
(4) Delete a series.
(5) Print series report - 2025
(6) Exit Application.
```

1.2. If the user selects to capture a new TV series, you are required to save all the information supplied by the user to memory. To achieve this, you may use arrays or array lists to save the series model. The series model should consist of the following:

```
public class SeriesModel {
    public String SeriesId;
    public String SeriesName;
    public String SeriesAge;
    public String SeriesNumberOfEpisodes;
}
```

Sample Capture Series Screen Shot

```
CAPTURE A NEW SERIES
*****
Enter the series id: 101
Enter the series name: Extreme Sports
Enter the series age restriction: 12
Enter the number of episodes for Extreme Sports: 10
Series processed successfully!!!
Enter (1) to launch menu or any other key to exit
```

- 1.3. If the user enters an incorrect age restriction, prompt the user to re-enter a valid age restriction. A valid age restriction is between 2 and 18. Only numbers are allowed to be supplied when entering an age restriction.

Sample Screen Shot of an invalid non-number age restriction.

```
Enter the series age restriction: Ten
You have entered a incorrect series age!!!
Please re-enter the series age >> |
```

Sample Screen Shot of an invalid age restriction.

```
Enter the series age restriction: 32
You have entered a incorrect series age!!!
Please re-enter the series age >> |
```

- 1.4. Once the entire new series has been captured, the user must be informed that the series details have been successfully saved.
- 1.5. The user must have the ability to search for a TV series. The user will select menu item two (2), which will prompt the user to enter a series id. If a series is found in the application, display the series details to the user. If no series is found display an error message to the user that no series data could be found.

Sample Series Search Screen Shot

```
Enter the series id to search: 101
-----
SERIES ID: 101
SERIES NAME: Extreme Sports
SERIES AGE RESTRICTION: 12
SERIES NUMBER OF EPISODES: 10
-----
Enter (1) to launch menu or any other key to exit
```

Sample Series Not Found Screen Shot

```
Enter the series id to search: 102
-----
Series with Series Id: 102 was not found!
-----
Enter (1) to launch menu or any other key to exit
```

- 1.6. The user must have the option to update any series that has been captured. The user must first enter in the series id to be updated. The user will have the functionality to update the series name, age restriction and the number of episodes.

Sample Series Update Screen Shot

```
Enter the series id to update: 101
Enter the series name: Extreme Sports 2025
Enter the age restriction: 10
Enter the number of episodes: 12
Enter (1) to launch menu or any other key to exit
```

- 1.7. The user must have the option to delete a series that has been captured. The user must first enter the series id to be deleted. The user must confirm whether they want to delete the series.

Sample Series Delete Screen Shot

```
Enter the series id to delete: 101
Are you sure you want to delete series 101 from the system? Yes (y) to delete.
Y
-----
Series with Series Id: 101 WAS deleted!
-----
Enter (1) to launch menu or any other key to exit
```

- 1.8. When the user selects to view the series report, display the report below, which is generated from the memory collection in your application.

Sample Report Screen Shot

```
Please select one of the following menu items:
(1) Capture a new series.
(2) Search for a series.
(3) Update series age restriction
(4) Delete a series.
(5) Print series report - 2025
(6) Exit Application.
5
Series 1
-----
SERIES ID: 101
SERIES NAME: Extreme Sports
SERIES AGE RESTRICTION: 12
NUMBER OF EPISODES: 10
-----
Series 2
-----
SERIES ID: 102
SERIES NAME: Bargain Hunters
SERIES AGE RESTRICTION: 10
NUMBER OF EPISODES: 10
-----
Series 3
-----
SERIES ID: 103
SERIES NAME: Home Cooking
SERIES AGE RESTRICTION: 10
NUMBER OF EPISODES: 20
-----
Enter (1) to launch menu or any other key to exit
```

Additional Requirements:

- 1.9. In your solution, you are required to create a class called Series, which will contain all your working methods.
- 1.10. This class will as a minimum, contain the following methods, but you are encouraged to add more methods:
- CaptureSeries();
 - SearchSeries();
 - UpdateSeries();
 - DeleteSeries();
 - SeriesReport();
 - ExitSeriesApplication();

1.11. Finally, create a main class to run your application.

Section B

(Marks: 25)

Your line manager has instructed you to write unit tests to determine the code coverage in the TV series management application. You are required to create a test package within the application you created in **Section A**, which will contain the required unit tests.

You are required to write the following unit tests:

Test Name	Test Purpose
TestSearchSeries()	To supply the series id to the series searching method. The test will determine that the correct series data has been returned.
TestSearchSeries_SeriesNotFound ()	To supply an incorrect series id to the series searching method. The test will determine that no series was found.
TestUpdateSeries()	To supply the Series id to the update series method. The test will determine that the series has been successfully updated.
TestDeleteSeries()	To supply the series id to the delete series method. The test will determine that the series has been successfully deleted.
TestDeleteSeries_SeriesNotFound()	To supply an incorrect series id to the delete series method. The test will determine that the series has not been deleted.
TestSeriesAgeRestriction_AgeValid()	To supply a valid series age restriction to the series age restriction method. The test will determine that the series age restriction is valid.
TestSeriesAgeRestriction_SeriesAgeInvalid()	To supply an invalid series age restriction to the series age restriction method. The test will determine that the series age is invalid.

Section B**Question 1****(Marks: 35)**

You have to design your own Java console application about any valid problem, which must be solved by your application. Your solution can include solving a business problem, a new idea or even a game. Your application must make use of concepts such as arrays, loops, inheritance, constructors and information hiding. The output must be shown in the form of a report using the console. You are also required to write unit tests to validate that your code achieves the desired result.

In your solution, make use of as many concepts and components dealt with in this course, but pay special attention to the following learning units:

- Learning Unit 1: Advanced arrays
- Learning Unit 2: Introduction to inheritance

END OF PAPER

Appendix A**Question 1: Section A - Mark Allocation**

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Variables declared and assigned. [5 Marks]	There is little to no attempt in the declaration and the assignment of variables.	The code attempts to include the declaration and assignment of variables.	The code contains excellent examples of the declaration and assignment of variables.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Menu and layout created [5 Marks]	There is little to no attempt to create a menu and layout in the application.	The code attempts to include a menu and layout in the application.	Excellent code provided for the menu and layout creation in the application.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Saving and capturing values to memory [5 Marks]	There is little to no attempt in the code to save and capture values to memory.	The code attempts to save and capture values to memory.	Excellent code provided for saving and capturing values to memory.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Searching for a series [5 Marks]	There is little to no attempt in the code to search for a series.	The code provided attempts to search for a series.	Excellent code provided for searching for a series and returns the correct result.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Deleting a series [5 Marks]	There is little to no attempt in the code for the deletion of a series.	The code provided attempts to delete a series.	Excellent code provided for the deletion of a series, which removes the item from memory.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Updating a series [5 Marks]	There is little to no attempt in the code for the updating of a series.	The code provided attempts to update a series.	Excellent code provided for the updating of a series, which updates the required memory items.
	0 – 1 Marks	2 – 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Series report generated [3 Marks]	There is little to no attempt in the code for the creation and displaying of the series data report.	The code provided attempts to create and display the series data report.	Excellent code provided for the creation and displaying of the series data report.
	0 Marks	1 - 2 Marks	3 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
A series class was created with working methods. [3 Marks]	There is little to no attempt in the creation of the series class with working methods.	Code provided attempts in the creation of the series class with working methods.	Excellent code provided for the creation of the series class with working methods.
	0 Marks	1-2 Marks	3 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Good programming practice and comments [4 Marks]	There is little to no attempt to meet good programming practice and comments provided.	Code provided attempts good programming practice and provides adequate comments.	Excellent code provided with good programming practice and comments provided.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Question 1: Section B - Mark Allocation

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Search Series unit test written with the correct test result [4 Marks]	There is little to no attempt in the creation of the Test Search Series unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Test Search Series unit test, with the desired test outcome achieved.	Excellent creation of the Test Search Series unit test and the desired test outcome achieved.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Search Series Not Found unit test written with the correct test result [3 Marks]	There is little to no attempt in the creation of the Test Search Series Not Found unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Test Search Series Not Found unit test, with the desired test outcome achieved.	Excellent creation of the Test Search Series Not Found unit test, and the desired test outcome was achieved.
	0 – 1 Marks	2 Marks	3 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Update Series unit test written with the correct test result [4 Marks]	There is little to no attempt in the creation of the Update Series unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Update Series unit test, with the desired test outcome achieved.	Excellent creation of the Update Series unit test and the desired test outcome achieved.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Delete Series unit test written with the correct test result [4 Marks]	There is little to no attempt in the creation of the Test Delete Series unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Test Delete Series unit test, with the desired test outcome achieved.	Excellent creation of the Test Delete Series unit test, and the desired test outcome was achieved.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Delete Series Not Found unit test written with the correct test result [3 Marks]	There is little to no attempt in the creation of the Test Delete Series Not Found unit test with the desired test outcome achieved.	An attempt was made in the creation of the Test Delete Series Not Found unit test with the desired test outcome achieved.	Excellent creation of the Test Delete Series Not Found unit test and the desired test outcome achieved.
	0 – 1 Marks	2 Marks	3 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Series Age Restriction: Age Valid unit test written with the correct test result [4 Marks]	There is little to no attempt in the creation of the Test Series Age Restriction Age Valid unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Test Series Age Restriction Age Valid unit test, with the desired test outcome achieved.	Excellent creation of the Test Series, Age Restriction, Age Valid unit test, and the desired test outcome achieved.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Test Series Age Restriction: Age Invalid unit test written with the correct test result [3 Marks]	There is little to no attempt in the creation of the Test Series Age Restriction Age Invalid unit test, with the desired test outcome not achieved.	An attempt was made in the creation of the Test Series Age Restriction Age Invalid unit test, with the desired test outcome achieved.	Excellent creation of the Test Series, Age Restriction, Age Invalid unit test, and the desired test outcome achieved.
	0 – 1 Marks	2 Marks	3 Marks
Mark Allocated			
Feedback			

Appendix B**Question 1 Mark Allocation**

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Variables Declared [3 Marks]	There is little to no attempt in the creation and declaration of variables.	An attempt was made in the creation and declaration of variables.	Excellent creation and declaration of variables.
	0 – 1 Marks	2 Marks	3 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
A console application created with an acceptable layout [4 Marks]	There is little to no attempt in the creation of a console application, and the creation of an acceptable layout.	An attempt was made in the creation of a console application and the creation of an acceptable layout	Excellent creation of a console application and an acceptable layout.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Concepts such as arrays, loops, inheritance, constructors and information hiding. [5 Marks]	There is little to no attempt in the creation of concepts such as arrays, loops, inheritance, constructors and information hiding.	An attempt was made in the creation of concepts such as arrays, loops, inheritance, constructors and information hiding.	Excellent creation of concepts such as arrays, loops, inheritance, constructors and information hiding.
	0 – 1 Marks	2 - 3 Marks	4 – 5 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Good programming practice and comments [4 Marks]	There is little to no attempt in good programming practice and creation of comments.	An attempt has been made for good programming practice and creation of comments.	Excellent programming practice and creation of comments.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Creative application with more than basic functionality and effort [4 Marks]	There is little to no creativity in the application, and no extra effort has been made.	An attempt has been made to make a creative application, and extra effort has been made.	Excellent application creativity and extra effort were made.
	0 – 1 Marks	2 - 3 Marks	3 – 4 Marks
Mark Allocated			
Feedback			

Marking Criteria	Does not meet the required standard	Meets the required standard	Exceeds the required standard
Unit tests are created to prove that the code functions as intended [15 Marks]	There is little to no attempt to create unit tests that prove that the code functions as intended.	An attempt has been made to create unit tests that prove that the code functions as intended.	Excellent creation of unit tests that prove that the code functions as intended.
	0 – 4 Marks	5 - 9 Marks	10 – 15 Marks
Mark Allocated			
Feedback			